

Homework Answer Key

The Bridge: Scratch → Python · Your First Python "About Me" Program — Type it, run it, fix errors, bring it Monday · ⌚ ~30 min

- PYTHON 3
- PRINT()
- FOR LOOP
- 3 TIERS
- DUE MON JUL 13

YOUR NAME

First Last

DATE COMPLETED

MM / DD

SECTION / PERIOD

WHERE TO TYPE YOUR CODE

Go to **playcode.io/python-playground** — free, browser-based, no install needed. File on left, output on right. Press ▶ to run. Can't get online? Write your code in the boxes below and bring it to class Monday — full credit!

PART 1 · CIRCLE YOUR TIER — ANSWER GUIDE (5 MIN)

All tiers do the **same required part** plus their own bonus. Award **2 pts** if:

- 4 facts planned with **real personal info** (not left blank or "N/A")
- Loop purpose circled or written: name 5× or count 1–5 or other valid choice
- Loop runs count **matches** the chosen range (name 5× → 5; count 1–5 → 5)

TIER A

Gr 1–3 / No Python yet · Required + loop bonus

TIER B

Gr 3–5 / Some Scratch · Required + loop + stretch

TIER C

Gr 6–9 / Python-Curious · Required + loop + two bonuses

⚠ Partial credit (1 pt): facts present but loop count blank or mismatched

PLAN BEFORE YOU TYPE — SAMPLE

What 4 facts will you `print()` about yourself?

Fact 1: My name is Jordan

Fact 2: I am 11 years old

Fact 3: I like playing basketball

Fact 4: My goal is to make a video game

Loop will: print my name 5×

Loop runs 5 times

The loop purpose and count must logically match. Name 5× → `range(5)` → 5 runs. ✓

PART 2 · REQUIRED — ALL TIERS · WRITE & RUN YOUR PROGRAM (15–20 MIN)

Sample full-credit program:

```
print("My name is Jordan")
print("I am 11 years old")
print("I like playing basketball")
print("My goal is to make a video game")
```

Grading checklist (all 4 = full 4 pts):

Exactly 4 `print()` lines present

Real personal info (not template placeholders)

All text in "double quotes"

Program runs or student documented fix

Error Check — Common fixes:

Missing quotes? → `NameError` — text treated as variable

Fix: `print("hello")` not `print(hello)`

Wrong parens? → `SyntaxError`

Fix: `print("hello")` both parens required

Capital P? → `NameError: Print is not defined`

Fix: `print` — all lowercase always

Error I got (model): `NameError: name 'Jordan' is not defined`

How I fixed it: Added "quotes" around the name

TIER A BONUS — ANSWERS (+2)

Add a **for loop** that prints your name 3 times.

Scratch: repeat 3 → say "name"

```
for i in range(3):
    print("Jordan")
```

TIER B BONUS — ANSWERS (+3)

Loop that **counts AND prints** your name. Add `#` comments.

```
# About Me
print("My name is Jordan")
print("I am 11 years old")
print("I like basketball")
print("My goal: make a game")
```

TIER C BONUS — ANSWERS (+4)

```
# Countdown
for i in range(10, 0, -1):
    print(i)
print("Blast off! 🚀")
```

Teach-Back: "print()" shows a message on screen. A for loop repeats code — you

Blank 1 → 3

Blank 2 → "Jordan" (student's name)

Output:

Jordan
Jordan
Jordan

Stuck? Ask a parent or sibling — that's fine!

```
# Counter loop
```

```
for i in range(1, 6):  
    print(i, "Jordan")
```

i on 3rd loop: 3 (range starts at 1)

Predicted output:

1 Jordan · 2 Jordan · 3 Jordan · 4 Jordan ·
5 Jordan

💡 range(1,6) gives 1,2,3,4,5 — stops **before** 6!

set how many times with range(). The body must be indented 4 spaces. It's like Scratch's repeat block."

range(0,10,3) gives: 0, 3, 6, 9

Starts at 0, adds 3, stops before 10.

Why does Python start at 0? Zero-based indexing — 5 items occupy positions 0,1,2,3,4 which is exactly 5 slots. Makes index math consistent.

PART 3 · RECORD & REFLECT — MODEL ANSWERS (5 MIN)

Copy your output below or take a screenshot to bring Monday.

Sample output (4-line program):
My name is Jordan
I am 11 years old
I like playing basketball
My goal is to make a video game

1. What does `print()` do?

`print()` outputs (displays) text or a variable's value to the screen/console.

2. What does the 4-space indent in a loop tell Python?

The 4-space indent tells Python the line is **inside the loop** — it runs on every iteration. Without it: `IndentationError`.

3. Did you get an error? What happened?

Any honest answer earns full credit:
"Yes — forgot quotes → `NameError` → added quotes and it worked."
or "No errors on first try."
No credit for blank only.

One thing that surprised me — sample answers:
"Python starts counting at 0, not 1!"
"One missing colon breaks everything — syntax is strict!"
"range(5) gives 5 numbers but the last is 4, not 5."

SCORING RUBRIC		
SECTION	WHAT TO LOOK FOR	PTS
Part 1 — Plan	4 facts planned; loop purpose & count filled in	2
Required — 4 prints	All 4 lines with real info; in quotes; no errors	4
Tier A Bonus	Loop written; blanks filled; output recorded	+2
Tier B Bonus	Loop typed; output predicted; <code>i</code> identified	+3
Tier C Bonus	Countdown built; teach-back written; investigate attempted	+4
Part 3 — Reflect	Output recorded; all 3 questions answered	2
Required total/ With bonus		8 / 16

NEXT SESSION — MON JUL 13

Session 9: Python Variables, `input()` & a Mad Libs Generator!
You'll learn how Python can ask you a question and remember your answer — the foundation of every interactive program. 🦋

Prepare: Think of a funny Mad Libs story topic (animals? food? school?). You'll need 4+ fill-in-the-blank words.

AI Bridge Note: Today's `print()` always gives the same output. What would make a program produce different output each time? → leads to `input()` — and to how AI models generate unique responses to every prompt.